

Validity Analysis: ARABICMMLU

Target context: MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Overall risk: **HIGH**

Deployment Context

Use case: Educational tutoring and supporting students studying with answers related to their curriculums in different subjects on the primary, middle, and high school, level as well as some university level subjects (management, economics, law, CS, political science, accounting)

Target population: Students (primary, middle, high school, and university) from Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	3	Moderate gaps	high
Input Content 	2	Significant gaps	high
Input Form 	4	Minor gaps	high
Output Ontology	3	Moderate gaps	high
Output Content 	2	Significant gaps	high
Output Form 	4	Minor gaps	high
Average	3.0		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

ArabicMMLU is the only large-scale natively-sourced Arabic exam benchmark available and is genuinely better-suited for MENA evaluation than translated MMLU alternatives [Q5, Q6]. For the deployment's MSA school-level STEM coverage and overall MCQ format alignment, it provides meaningful and trustworthy signal. However, for the deployment's eight-country curriculum-alignment requirement, ArabicMMLU has serious validity limitations that the dataset analysis confirms empirically rather than merely speculatively. The most acute concerns are content-level (IC) and label-correctness (OC): Jordanian-curriculum civics, history, social science, and economics content is encoded as universal ground truth across school

levels [DATASET-D1–D7, D20–D22, D25, D30]; university-level non-CS subjects come almost exclusively from a single Egyptian university [DATASET-D39, D40]; Law (Professional) is exclusively Moroccan procedural law in the sample [DATASET-D8–D10, D34]; Kuwait has no observed examples and no documented annotators [Q31, Q96]; Palestinian content (top-three contributor by volume) was verified without Palestinian annotators [Q96, WEB-1]. The flat label schema [Q27] cannot encode jurisdiction-specific correctness despite the deployment user explicitly requiring per-country legal tailoring [elicitation A2]. University-level coverage is also structurally thin (~6.1% [Q71]) for a deployment serving university students.

ID	Evidence
Q5	"Although large language models (LLMs) such as GPT-3.5 (Ouyang et al., 2022), BLOOMZ (Muennighoff et al., 2022), and Jais (Sengupta et al., 2023) have been pretrained with substantial coverage of Modern Standard Arabic (MSA), their reasoning and knowledge assessments are primarily conducted using datasets translated from English to Arabic (Sengupta et al., 2023; Huang et al., 2023), which means there is limited capacity to evaluate content specific to Arabic." (p.1)
Q6	"This reliance on translation systems not only demonstrates an Anglocentric approach (Ramesh et al., 2023; Talat et al., 2022) but also potentially introduces errors and biases." (p.1)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)
D1	What are the manifestations of change in modern Jordanian society? — country-specific Jordanian civics
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D8	ىوعدل رهوج يف مكحل رودص دعب ال عوفد وأ ضراع عازن يف ةرداصل وأ ةيديهمهملت ماكحلأ فانئتسا لبقي ال — Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D2	"The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D3	"ب وه ةفاك ةلودل نوؤش لىل ةماعل ةيألول بحاص سللمج" with answer "ءرازلو سللمج" — "The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom
D5	"لكلمل ةلالج دهع يف 1952 ماع روتسد رص" with options referring to Jordanian kings — "The 1952 constitution was issued during the reign of King [Talat bin Abdullah]" — Jordanian royal history
D6	وهو يزكزل كنبلل ينورتكلال عقومل قيرط نع ىوكشل لاسرا www.cbj.gov.jo — "Complaints to the Central Bank [of Jordan] via www.cbj.gov.jo " — explicitly references Jordan's central bank
D7	ةمسرل ةيلاامل تامدخ لىل لوصولو نوعيطتسي ال نذل ينيندرأل ةبسن 0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D20	ةيطلأ نيدمل ىفشتم :اهنم...تاسسؤمل نم ديدعل ةيموكحل ةيكلملل ةيطلأ تامدخل لمشت — "Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services
D21	ةنس همكح ةدمو 1952م: يف لكذو ةيحص بابسلأ هتيالو تهنتا نيسحل نبال لال كللمل — "King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school
D22	ةرجنع يف لبجل ةديس ةسي نك دجوت — "Lady of the Mountain Church is in Anjara [Jordan]" — Jordanian religious geography
D9	عافدلأ ةفل كللمل ةيموكحل ةطلسلأو لدعل ربزو نم رارقب...نينكي كردلل ةيئاضقلأ طشرلل طباض ةقص لىوخت نكمي — Moroccan judicial police powers — references "يكللمل كردلأ" (Royal Gendarmerie of Morocco)
D10	ةحيحص ةبوجلأ عيجم: مكحل رودص لبقي نيدمل فرطل لزانن اذإ — Moroccan civil party withdrawal in criminal proceedings
D25	بادتئال كف دويقي نم دالبل رحت: ١٩٢٨ ماع هي ناطي ربل هي ندرأل هدهاعلمل سيئرل فدهل ناك — "The main goal of the 1928 Jordanian-British treaty was liberation from Mandate restrictions" — Jordanian national history

D30	مريداو...مكةرتشم قرا دلا اهل ع فرشت ندرألا يف عي عي ب ط م حم — "Wadi Rum is a Jordanian nature reserve under joint management" — Jordan-specific geography fact
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF

Practical Guidance

What This Benchmark Measures

ArabicMMLU can validly measure: (a) MSA reading comprehension and Arabic grammar at school level (input_form score 4); (b) MCQ-format scoring alignment for the deployment's primary use case (output_form score 4); (c) STEM factual recall across all eight countries since STEM content is curriculum-neutral [DATASET-D24]; (d) Jordanian school-curriculum knowledge specifically (the dominant cohort); (e) some Egyptian university accounting/economics/management/political science alignment, subject to OCR caveats; (f) Moroccan procedural law specifically. The benchmark also supports rough comparative evaluation of model robustness across prompt/output configurations [Q47, Q105–Q110].

Construct Depth

Construct depth is shallow for the deployment's stated needs. The benchmark probes factual recall and modest reasoning at primary/middle/high school levels well, but university-level depth is thin (~6.1% of items, ~889 questions across all subjects [Q71]). Country-specific construct depth is highly uneven: Jordan is over-probed, Kuwait is essentially un-probed, Morocco is probed only in law and a few Arabic items, Palestine is probed in STEM but not verified by Palestinian experts. Madhab-specific Islamic studies depth is undocumented [gap_id 7]. Jurisdiction-specific legal depth is single-jurisdiction in the sample [DATASET-D34]. Construct depth strengths are concentrated in dimensions input_form and output_form; depth weaknesses are concentrated in input_content and output_content — the dimensions the elicitation marked HIGH priority.

What Else You Need

For a defensible deployment, ArabicMMLU should be supplemented with: (1) per-country re-annotation of civics, history, social science, economics, and law subsets by curriculum experts from Morocco, Kuwait, Palestine, UAE (addresses input_content and output_content gaps); (2) ArabLegalEval [WEB-3] for Saudi-specific law and MizanQA [WEB-4] for additional Moroccan law coverage, with a custom evaluation built for Egyptian/UAE/Jordanian/Lebanese/Kuwaiti/Palestinian legal jurisdictions which remain unfilled; (3) AraSTEM [WEB-2] for richer STEM coverage at university level (addresses input_ontology thinness); (4) a French-Arabic bilingual evaluation for Moroccan university science (no existing benchmark, must be built [gap_id 6]); (5) madhab-stratified Islamic studies evaluation built with Maliki and Hanbali curriculum experts [gap_id 7]; (6) cleaning or replacement of OCR-corrupted Management and Political Science (University) items [DATASET-D14, D15, D29].

ID	Evidence
Q105	"Zero-shot LLM performance (% accuracy) with Arabic prompt and Arabic alphabetic output, combined across subject groups." (p.16)
Q106	"Average" means the average across all questions in ArabicMMLU." (p.16)
Q107	"Table 9 presents results organized by subject groups: STEM, Social Science, Humanities, Arabic Language, and Other." (p.16)
Q108	"Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups." (p.17)
Q109	"Table 10: Zero-shot LLM performance (% accuracy) with Arabic prompt and English alphabetic output, combined across subject groups." (p.17)
Q110	"Zero-shot LLM performance (% accuracy) with English prompt and Arabic alphabetic output, combined across subject groups." (p.18)
Q47	"We initially conducted experiments with four settings: (1) Arabic prompt and Arabic alphabetic output, (2) Arabic prompt and English (i.e. Latin script) alphabetic output, (3) English prompt and Arabic alphabetic output, and (4) English prompt and English alphabetic output." (p.5)

D24	إذا جت ن اذإ Calvin cycle question — أهيتبث مت يت التا (CO2) تائيزج ددع مكف نفالك ةقلح نع زوكولغ لتائيزج (3) جتن اذإ Palestine-sourced biology; universal factual content
Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
WEB-3	ArabLegalEval reuses ArabicMMLU law subsets without jurisdiction tagging — confirms multi-jurisdiction gap is structural (https://arxiv.org/abs/2408.07983)
WEB-4	Arabic LLM benchmark survey confirms no French-Arabic bilingual benchmark for Moroccan higher education exists (https://arxiv.org/abs/2510.13430)
WEB-2	https://arxiv.org/abs/2501.00559
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source ي إ ل ري تذت...ةفتل تخسلا تي رادلإ تايهت تدسل او صاخ تلألأ ني تب طتل دل تا عتي هت عت جرد
D15	Severely OCR-corrupted management question with unreadable answer options ت ح ل ...م ح رخر لا ...ما دخ ست لإ س ئا طط خ لا ي ل غ ل اث م د عي لا
D29	Severely OCR-distorted Arabic text about conditions for an Islamic ruler — from Egyptian source م ار رسال ار ول دل أر ف مكار لا أر ف ار رفاو ت بر ررت لأرس اسالا طورر رشل ار نرم